

Cross-clausal movement and its limits¹

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Abstract

Constraints on movement are of central concern to syntactic theory. Recent work has focused on the relationship between the height of the landing site and locality restrictions. Keine (2020) argues that the higher the landing site the fewer locality restrictions are imposed on the movement (“higher is freer”). In contrast Deal (2017) argues that a lower landing site can lead to movement being less restricted due to Delayed Opacity, i.e. “lower is freer”. In this paper, I show that neither analysis makes the right predictions for movement out of finite clauses in Kalaallisut (Inuit, Greenland). I argue that, in Kalaallisut, it is not the height of the landing site that determines whether movement across a finite clause boundary is possible, rather it is the nature of the features driving the movement. In particular, ϕ -probing unlocks the embedded CP allowing for extraction from within. Focus probing does not, which accounts for the clause-bounded nature of focus fronting. In addition to deriving the differing locality profiles of hyperraising and focus fronting, the analysis of focus fronting generalizes to *wh*-movement and also accounts for apparent long-distance focus movement. The final segment of the paper situates the Kalaallisut pattern in a preliminary cross-linguistic typology of height-locality correlations.

Keywords: phase unlocking, hyperraising, Height-Locality Connection, focus movement, Q-particles, Inuit

1 Introduction

Many languages show patterns of DISPLACEMENT, in which a syntactic element appears in one position but is connected to another position as well. In (1), *who* appears in clause-initial position, but is also related to the post-verbal object position of *visit*.

(1) **Who** should we visit ____?

A massive body of work has argued that such displacement patterns involve MOVEMENT. A particular movement can be characterized by three specifications:

- ① what element moves?
- ② where does it move to (landing site)?
- ③ what material can it cross on its way to its landing site (locality)?

In (1), a *wh*-phrase moves to clause-initial position (specifier of CP) crossing a finite TP boundary [_{TP} *we visit*]. Much work has argued that these specifications are not independent of each other:²

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²All of these are areas of active research and disagreements. For instance, Matushansky (2006) and dos Santos (2025) argue for head movement to specifier position and Harizanov and Gribanova (2019) argue that syntactic head movement can cross head positions in violation of the Head Movement Constraint.

- connecting ① to ②:

STRUCTURE PRESERVATION: phrases move to specifier positions and heads move to head positions (e.g. Emonds 1970).

- connecting ① to ③:

HEAD MOVEMENT CONSTRAINT: movement of heads is subject to a more stringent locality condition than movement of phrases (e.g. Travis 1984).

- connecting ② to ③:

HEIGHT LOCALITY CONNECTION: movement to a low landing site is subject to more stringent locality restrictions than movement to a high landing site (e.g. Keine 2020).

The existence and nature of such connections are of central concern to syntactic theory, since they have the potential to reveal the mechanism of movement and the principles that constrain it. Here we will be concerned with the connection between ② and ③, specifically asking the following question: considering movement across a finite clause boundary to different landing sites, does the height of the landing site correlate with whether this movement is possible or not? This is by no means a new question. A celebrated answer from English is based on the contrast between (2) and (3):

- (2) [_{CP} **Who** do you think [_{CP} ___ will visit my mother]]? *wh*-MOVEMENT
 (3) *[[_{TP} **Delia** seems [_{CP} ___ will visit my mother]] RAISING

In this pair of movement structures, the height of the landing site correlates with the possibility of crossing a finite clause boundary. *Wh*-movement in (2) targets a high position, namely the specifier of CP, and can cross a finite clause boundary. Raising in (3) targets a lower position, namely specifier of TP, and cannot cross a finite clause boundary. Keine (2020) generalizes this into a universal Height-Locality Connection: the higher the landing site of the movement, the less restricted the movement is in the boundaries it can cross. (See also Williams 2003, Abels 2012, Müller 2014, Wurmbrand 2015.) Against this back-drop, we argue that Kalaallisut (Inuit, Greenland) shows the opposite pattern of the English one above: movement that targets a low position can cross a finite clause-boundary (hyperraising to object), but movement that targets a high position cannot cross a finite clause boundary (focus fronting and *wh*-movement). This state of affairs is shown in the table below.

- (4) EXPECTED: HIGH LESS RESTRICTED THAN LOW (English)

landing site	cross-clausal
high	✓
low	✗

- (5) UNEXPECTED: LOW LESS RESTRICTED THAN HIGH (Kalaallisut)

landing site	cross-clausal
high	✗
low	✓

The Kalaallisut pattern suggests that height-locality connections are not unidirectional once we look cross-linguistically and therefore casts doubt on Keine's Height-Locality Connection as a universal. Instead of the height of the landing site determining whether cross-clausal movement is possible, we will argue

that it is the nature of the movement, specifically which feature triggers it, that explains the Kalaallisut pattern. The analysis combines two ideas: i) PHASE UNLOCKING (e.g. Rackowski and Richards 2005; van Urk and Richards 2015; Branan 2018; Davis and Branan 2019; Lee and Yip 2024; Little 2025)³ and ii) CONJUNCTIVE PROBING (e.g. Van Urk 2015, Scott 2021, Lohninger et al. 2022). This combination is prefigured in Lee and Yip (2024)’s analysis of hyperraising to subject in Cantonese and Vietnamese, which will be discussed in section 7. However, Lee and Yip (2024) do not consider the contrast between hyperraising and A-bar movement that is in focus here.

The paper is organized as follows. In section 2, we provide background on Kalaallisut and our research collaboration. In section 3, we establish the locality profiles of hyperraising and focus fronting. The analysis is developed in section 4, and section 5 confirms two predictions it makes. Section 6 considers two alternative analyses: i) Delayed Opacity (Deal 2017) and ii) Horizons (Keine 2020) and contrasts them with the phase-unlocking analysis in terms of the predictions they make for *wh*-movement. Section 7 returns to phase unlocking and argues that there are two kinds—relativized and generalized—and discusses the theoretical implications of this finding. Finally, section 8 summarizes our conclusions and lays out the broader cross-linguistic landscape.

2 Background

This work represents a collaboration between first author linguist, second author Kalaallisut speaker, and third author Kalaallisut speaker. The collaboration between the first and second author began in the summer of 2022 and the third author joined the project in May of 2024. The data discussed below come from one-on-one weekly Zoom elicitation sessions, which also included discussion of linguistic questions of interest. All elicitation recordings are archived in the California Language Archive. To contextualize the language of the second and third authors, brief autobiographies are included (omitted here for anonymity).

Language ecology Kalaallisut is a member of the Inuit-Yupik-Unangan language family, which is spoken from Eastern Russia across Alaska and Canada to Greenland. It is the sole official language of Greenland, with around 50,000 speakers in Greenland and another 17,000 in Denmark (Naja Blytman Trondhjem p.c.). The language is acquired by children in the home and taught in school. Danish is/was a colonial power in Greenland and Danish language has a big presence, especially in administration. English has a growing presence.

Basic linguistic features Key linguistic properties of Kalaallisut are illustrated in (6) through (8).^{4,5}

- (6) *pro* **Kaagi**-sor-uit *pro* *pro* ajori-ssu-akkit.
2SG.ABS cake-consume-2SG.CON 1SG.ERG 2SG.ABS angry-FUT-1SG>2SG.IND
If you eat cake, I'll be angry with you. Second author ANM-1
- (7) Naja-**p** Juuna ikior-**paa**.
Naja-ERG Juuna.ABS help-3SG>3SG.IND
Naja helped Juuna. Second author AFX-1

³Phase unlocking is referred to as phase deactivation by Lee and Yip (2024).

⁴Each example is annotated with the name of the speaker and an alpha-numerical code identifying it in the first author's notebooks.

⁵Glossing conventions: ABS = absolutive; ALL = allative case; CON = conditional mood; CONJ = conjunction; ERG = ergative; FOC = focus; FUT = future; IND = indicative mood; INT = interrogative mood; PART = participial mood. 3SG>3SG.IND means third person singular subject acting on third person singular object in indicative mood.

- (8) Juuna angerla-jaar-**poq**.
 Juuna.ABS come.home-early-3SG.IND
 Juuna came home early.

Second author AFX-2

Example (6) illustrates two key polysynthetic properties: noun incorporation (Sadock 1980) and pro drop (Yuan 2018:154). It further shows that both intransitive and transitive agreement morphology is fused with mood morphology: dependent conditional mood in the embedded clause and independent indicative mood in the matrix clause. Of eight moods only three will be relevant in what follows: indicative, interrogative and participial. Example (7) illustrates the unmarked SOV order. SOV order is accepted in any context in elicitation, it is the order volunteered in out-of-the blue contexts, and it is the dominant word order in texts (Fortescue 1993). (See also Fortescue 1984:93ff, and Kleinschmidt 1851/1968:98–101 for arguments that SOV is the unmarked order.) Together with (8), (7) shows that case alignment is ergative-absolutive. The transitive subject is overtly marked with the ergative suffix *-p*, whereas transitive objects and intransitive subjects are morphologically unmarked, as is common for absolutive case. There is no morphological ergative/absolutive case distinction in pronouns and plural nouns, but we gloss them with case based on their grammatical function. Complement clauses typically appear after the matrix verb and typically trigger 3SG object agreement on the matrix verb (Fortescue 1984:34):

- (9) Nalunngi-**lara** ajugaa-ssa-sutit.
 know-1SG>3SG.IND win-FUT-2SG.PART
 I know you are going to win.

Third author BZP-GS-1

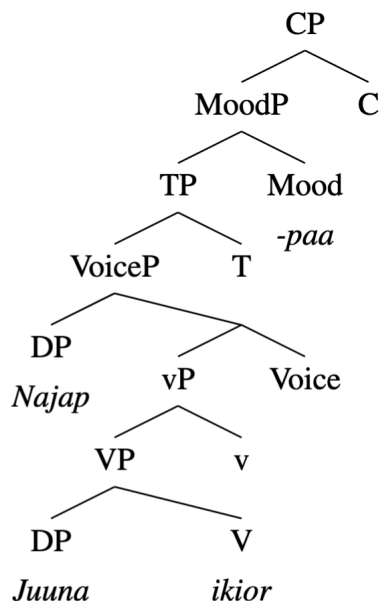
There are however a dozen intransitive embedding verbs, including *isumalior*- ‘think’ in (10), which agree only with the matrix subject.

- (10) Isumalior-**punga** ajugaa-ssa-sutit.
 think-1SG.IND win-FUT-2SG.PART
 I think you are going to win.

Third author CEH-GS-1

Based on the discussion above, we propose the basic clause structure in (11), illustrated with the sentence in (7).

(11)



Compton (2018) collapses Mood and C, whereas Yuan (2018:32) assumes that MoodP is dominated by subject and object agreement projections. As far as we can tell, nothing of consequence hinges on this choice. The points we make could be made under either of the alternative structures.

3 Two movement processes in Kalaallisut

Against the backdrop of Kalaallisut’s underlying SOV order, we now consider two movement processes, hyperraising to object and focus fronting. What is of primary interest is that these movements show different locality profiles. Specifically, hyperraising can cross a finite clause boundary and focus movement cannot. For legibility, hyperraising is indicated by boxing in the following examples and focus fronted phrases are bolded.

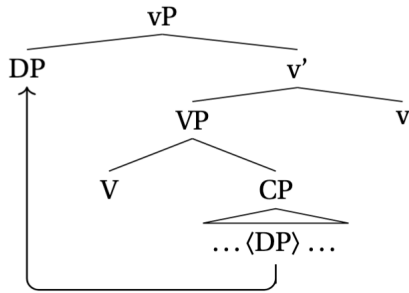
3.1 Hyperraising

As noted in Fortescue (1984:38) and Sadock (2003:32), and discussed at length in Mikkelsen and Thrane (2023), Kalaallisut allows raising to object out of a finite clause, i.e. hyperraising to object:⁶

- (12) Eqqaama-vara [illit angerla-jaar-tutit.]
remember-1SG>3SG.IND 2SG.ABS come.home-early-2SG.PART
I remember that you came home early. BASELINE – Third author CCB-GS-5
- (13) Illit eqqaama-vakkit [2SG.ABS remember-1SG>2SG.IND leave-early-2SG.PART]
I remember that you came home early. HYPERRAISING – Second author ACU-7

Mikkelsen and Thrane (2023) show that the configuration in (13) involves movement of *illit* from the lower clause into the matrix clause where it triggers object agreement on the matrix verb. They provide evidence from island and locality effects that this is movement, though we won’t repeat their data. Of interest here is that this is movement across a finite clause boundary to a low position:

- (14) HYPERRAISING OUT OF A FINITE CP



The finiteness of the embedded clause is evidenced by its obligatory mood-agreement morphology—compare (15) to (16)—and its ability to express tense independently of the tense of the matrix clause (17):

⁶Other languages that have been argued to have hyperraising to object include Janitzio P’urhépecha, (Zyman 2017), Japanese (Tanaka 2002), Kipsigis (Jake and Odden 1979), Korean (Yoon 2007), Mongolian (Fong 2018), Nez Perce (Deal 2017, covertly), Romanian (Alboiu and Hill 2016), Spanish (Herbeck 2020), and Zulu (Halpert and Zeller 2015).

- (15) *Illit eqqaama-vakkit [__ angerla-jaar].
 [2SG].ABS remember-1SG>[2SG].IND come.home-early
 I remember that you came home early. Second author BZS-3
- (16) Illit eqqaama-vakkit [__ angerla-jaar-**tutit**].
 [2SG].ABS remember-1SG>[2SG].IND leave-early-2SG.PART
 I remember that you came home early. HYPERRAISING – Second author ACU-7
- (17) Illit ilimaga-akkit [__ ajugaa-**ssa**-sutit].
 [2SG].ABS expect-1SG>[2SG].IND win-FUT-2SG.PART
 I expect that you will win. Third author BZZ-GS-4

These embedded clauses are therefore at least MoodPs. We will go one step further and assume that they are CPs, based on the fact that *wh*-movement can take place within such embedded clauses, suggesting the presence of a specifier of CP position (Keine 2020:57).

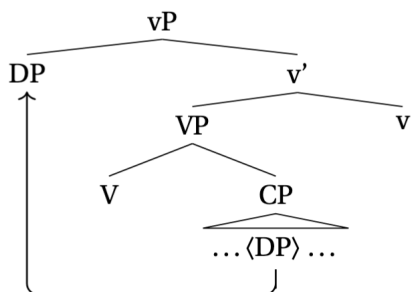
- (18) Nalu-akka [kikkut_i Naja-p ___i ikior-ai].
 not.know-1SG>3PL.IND who.PL.ABS Naja-ERG help-3SG>3PL.PART
 I don't know who all Naja helped. Second author AGH-6

The correlation of finiteness with the presence of a CP is in accord with most analyses of hyperraising (e.g. Halpert 2019, Martins and Nunes 2009, Fong 2018, Deal 2017), though Jenks (2025) argues that Moro hyperraising is out of a finite TP. The exact landing site of hyperraising is a little harder to pinpoint, but crucially it is lower than the landing site of focus movement. This is demonstrated in the examples in (19) and (20).

- (19) **Uanga=ana** meeqa-t eqqaama-gikka angerla-jaar-tut
 1SG.ERG=FOC.SG child-[PL].ABS remember-1SG>3[PL].PART come.home-early-3PL.PART
 It is me that remembers that the children came home early. Third author BZO-GS-1
- (20) *Meeqa-t **uanga=ana** eqqaama-gikka angerla-jaar-tut
 child-[PL].ABS 1SG.ERG=FOC.SG remember-1SG>3[PL].PART come.home-early-3PL.PART
 It is me that remembers that the children came home early. Third author BZO-GS-2

In the well-formed (19) the hyperraised object *meeqqat* occurs to the right, i.e. lower than the focus fronted DP. The ill-formedness of (20) shows that hyperraising cannot target a position above a focus fronted phrase. For concreteness we take the landing site of hyperraising to be the specifier of vP, as in (21), though all that matters for assessing the Height-Locality Connection is that it is below the landing site of focus fronting (= specifier of CP).

- (21) HYPERRAISING TO SPEC-VP



We have now established the locality profile of Kalaallisut hyperraising, namely that it can cross a finite clause boundary. This is surprising from the perspective of the Height Locality Connection, according to which movement to a low position is more restricted than movement to a high position.

(22)

	landing site	cross clausal
Expected	low	X
Kalaallisut	low	✓

3.2 Focus fronting

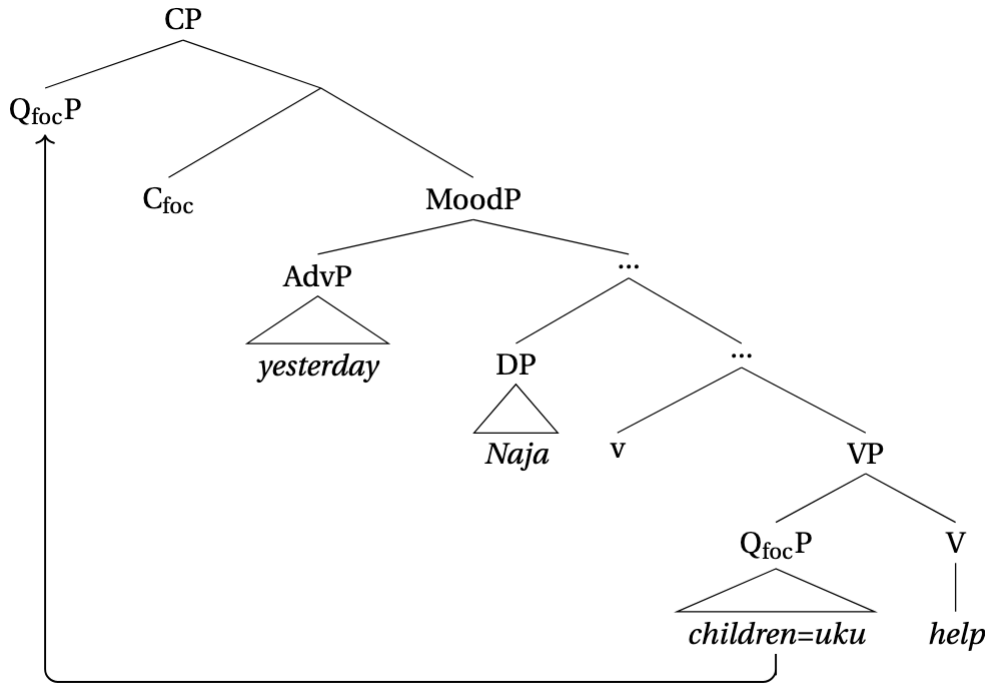
Kalaallisut has a focus construction in which the focused element appears in initial position followed by a focus clitic (=una for singular foci, =uku for plural foci) and the verb is realized in the participial mood:^{7,8}

- (23) **Meeqqa-n=uku** ippassaq Naja-p ikior-ai.
 child-PL.ABS=FOC.PL yesterday Naja-ERG help-3SG>3PL.PART
 It was the children that Naja helped yesterday.

Second author AOX-2

This construction has been noted and described in previous literature (Fortescue 1984:74ff, 184ff, Molich 2022:100-107), but, as far as we know, no theoretically oriented analysis has been developed. To begin to fill this gap we start by establishing four analytic properties of the construction. First, focus fronting is derived by movement. Second, focus fronting is clause bounded. Third, =una/=uku are focus Q-particles in the sense of Cable (2010), notated Q_{foc} . Q_{foc} takes the focused phrase as their complement and projects a $Q_{\text{foc}}\text{P}$. Finally, there is a dedicated focus C head that attracts $Q_{\text{foc}}\text{P}$ to its specifier. The resulting derivation for (23) is sketched in (24).

(24)



⁷=una and =uku trigger nasalization or lenition of a preceding consonant.

⁸To refer to the information structural status of a phrase I use ‘focus’ or ‘focused’. I gloss =una FOC and =uku FOC.PL. The morpho-syntactic feature corresponding to focus is notated FOC and used in probe specifications and Vocabulary Items. Finally, I use subscripted ‘foc’ to designate a subcategory. So C_{foc} stands for a focus C head.

The following sections establish each of these properties in turn. The details of the derivation are provided in section 4.4.

3.2.1 Focus fronting is movement

Evidence that focus fronting is movement comes from reconstruction and island effects.

Reconstruction effects If focus fronting is a species of A-bar movement, we expect it to exhibit reconstruction effects. That is, the moved element should give rise to syntactic and/or semantic effects associated with its base position. We illustrate with anaphor binding in (25)–(27). Kalaallisut possessive inflection distinguishes anaphoric third person possessors (glossed 4th person) from non-anaphoric 3rd person possessors (glossed 3rd person). Anaphoric possessors must be bound by a structurally higher subject, as seen in the contrast between the (25) and (26):⁹

(25) 4TH ANAPHORIC POSSESSOR INSIDE OBJECT AND BOUND BY SUBJECT

Hansi_i-p [pro_i qimmi-ni] maqaasi-vaa.
 Hansi-ERG 4SG.ERG dog-4SG>3SG.ABS miss-3SG>3SG.IND

Hans_i missed his_i dog.

Second author AQF-1

(26) 4TH ANAPHORIC POSSESSOR INSIDE SUBJECT AND NOT BOUND BY SUBJECT

*[pro_i qimmi-mi] Hansi_i maqaasi-vaa.
 4SG.ERG dog-4SG>3SG.ERG Hansi.ABS miss-3SG>3SG.IND

Intended: His_i dog missed Hansi_i.

Third author BNZ-GS-3

Now we observe that a possessive object DP with a 4th person possessor can be focus fronted and still be bound by the structurally lower subject (27):

(27) FOCUS FRONTING RECONSTRUCTS FOR ANAPHOR BINDING

[pro_i qimmi-ni]=una Hansi-p ____ magaasi-gaa.
 4SG.ERG dog-4SG>3SG.ABS=FOC Hansi-ERG miss-3SG>3SG.PART

It is his_i dog that Hansi_i missed.

Second author AQF-3

If focus fronting is movement, we can understand the grammaticality of (27) as reconstruction: for the purposes of anaphor binding, the possessive object DP (*qimmi-ni=una*) behaves as if it is in its base position below the subject (*Hansi-p*).

⁹4SG>3SG.ABS means an absolutive possessive DP with a 4SG possessor and a 3SG possessum. Possessors show ergative case marking when case is visible.

Island effects Since focus fronting is clause bounded, clause-sized islands like adjunct clauses, *wh*-complement clauses and sentential subjects are irrelevant. Instead we test for movement using the Coordinate Structure Constraint (28) and the Left Branch Condition (29). As (28) shows a focus marked DP cannot be extracted from a coordinate structure. This holds whether there is narrow focus on the extracted conjunct or on the entire coordination. Similarly (29) shows that a focus marked possessor cannot be extracted from a possessive DP. Again, this holds whether the focus is narrowly on the possessor or on the entire possessive DP.¹⁰

(28) EXTRACTION OUT OF COORDINATE STRUCTURE JUDGED UNGRAMMATICAL

***Illin=una_i** ippassaq [___i uanga=lu] angerla-jaar-tugut.
2SG.ABS=FOC yesterday 1SG.ABS=AND come.home-early-1PL.PART

Intended: It was you and me that came home early yesterday.

Second author AQD-3

(29) EXTRACTION OF POSSESSOR JUDGED UNGRAMMATICAL

***Meeqqa-n=una_i** Naja-p [___i qimmi-at] nani-gaa.
child-ERG.PL=FOC Naja-ERG dog-3PL>3SG.ABS find-3SG>3SG.PART

Intended: It was the children's dog Naja found.

Third author CGG-GS-4

This concludes the argument that focus fronting is derived by movement. Next we turn to the locality constraints on this movement.

3.2.2 Locality of focus fronting

Focus fronting cannot cross a finite clause boundary.¹¹ This restriction is illustrated for intransitive embedding verbs like *sinnattor*- 'dream' and *oqar*- 'say' in (30) and (31).

(30) ***Meeqqa-n=uku** sinnattor-tunga [Naja-p __ ikior-ai].
child-PL.ABS=FOC.PL dream-1SG.PART Naja-ERG help-3SG>3PL.PART

Intended: It's the children that I dreamt Naja helped.

Second author CAU-4

(31) ***Meeqqa-n=uku** Naja oqar-toq [__ iipili-t tillik-kaat].
child-PL.ERG=FOC.PL Naja.ABS say-3SG.PART apple-PL.ABS steal-3PL>3PL.PART

Intended: It is the children Naja said stole the apples.

Third author CDR-GS-2

This locality restriction also holds for transitive embedding verbs like *ilimagi*- 'expect' and *qulari*- 'doubt' in (32) and (33):

(32) ***Meeqqa-n=uku** ilimagi-giga [illit __ ikior-itit].
child-PL.ABS=FOC.PL expect-1SG>3SG.PART 2SG.ERG help-2SG>3PL.PART

Intended: It is the children that I expect you to help.

Second author CCL-1

¹⁰For reasons we don't currently understand, ungrammaticality judgments are very robust for extraction across an argument, but somewhat mixed for extraction across an adverb. The same is true for other types of extraction, specifically hyperraising, *wh*-movement, and scrambling.

¹¹This generalization is robust for transitive verbs. For the intransitive embedding verb *isumali*- 'think', the second author sometimes deems cross-clausal movement acceptable. In contrast, cross-clausal focus movement across other intransitive embedding verbs including *oqar*- 'say', *sinnattor*- 'dream', *tusar*- 'hear' and *allap*- 'write' are judged ungrammatical. At present we have no explanation for the exceptional behavior of *isumali*- 'think' in the second author's grammar.

- (33) ***Meeqqa-n=uku** qulari-giga [__ iipili-t tillik-kaat].
 child-PL.ERG=FOC.PL doubt-1SG>3SG.PART apple-PL.ABS steal-3PL>3PL.PART

Intended: It is the children that I doubt stole the apples.

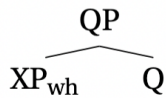
Third author CDQ-GS-2

In all of these examples the focus fronted phrase originates inside the embedded clause and surfaces in the main clause. As these examples show such structures are ungrammatical, demonstrating that focus fronting is clause bound.

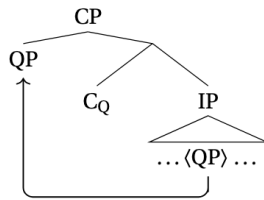
3.2.3 =una/=uku are focus Q-particles

Cable (2010) argues that so-called Q particles serve a mediating role in *wh*-questions. The Q-particle takes the *wh*-phrase as its complement (34) and the interrogative C, C_Q, attracts the entire QP to its specifier, giving the appearance of movement of the *wh*-phrase (35).

- (34) QP SHELL AROUND WH-PHRASE

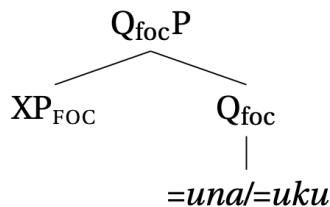


- (35) QP MOVEMENT

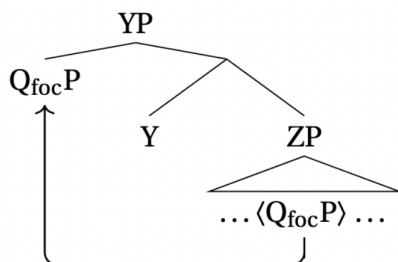


Following a suggestion in Cable (2010:200-205), We extend this general line of analysis to Kalaallisut focus fronting. Specifically, we propose that =una/=uku are Q-particles of the focus kind (Q_{foc}) that take the focused phrase as its complement (36). It is then the Q_{foc}P that moves, giving the appearance of focus fronting (37).

- (36) FOCUSED XP INSIDE Q_{foc}P SHELL



- (37) Q_{foc}P MOVEMENT



The XP sister of Q_{foc} in (36) can be instantiated by phrases of any category capable of being focused. This is shown for a DP in (38), a CP in (39), APs in (40) and (41), a numeral in (42) and adverbials in (43) and (44).¹²

- (38) **Meeqqa-n=uku** uanga ikior-ikka.
child-PL.ABS=FOC 1SG.ERG help-1SG>3PL.PART
It was the children I helped.’ Second author AMD-4
- (39) **Angerla-jaar-tutin=una** uanga eqqaama-giga.
come.home-early-2SG.PART=FOC 1SG.ERG remember-1SG>3SG.PART
It was that you came home early that I remember. Third author CEH-GS-5
- (40) **Ilu moortu-ming=una** Naja politikeri-sior-toq.
honest-INST=FOC Naja.ABS politician-see-3SG.PART
It is an honest politician that Naja is looking for. Second author CEE-3
- (41) **Aapaluttuming=una** illor-a qalipak-kiga.
red=FOC house-1SG>3SG.ABS paint-1SG>3SG.PART
It was with red that I painted my house. Third author CEI-GS-3
- (42) **Qulining=uku** pani-ga ukio-qar-toq.
ten=FOC.PL daughter-1SG>3SG.ABS year-have-3SG.PART
It is 10 that my daughter is. Second author CEF-2
- (43) **Ippassar=una** mee qqa-t Naja ikior-aat.
yesterday=FOC child-PL.ERG Naja.ABS help-3PL>3SG.PART
It was yesterday that the children helped Naja. Second author AOV-1
- (44) “**Soor=una** taama=ilior-tutit kuukkuuriaq” aapakaaq aperi-poq.
why=FOC thus=do-2SG.PART crocodile monkey ask-3SG.IND
“Why is it that you’re doing this, Crocodile?” asked the monkey.
Aapakaaq kuukkuuriarlu “The monkey and the crocodile”

Finally we turn to the landing site of focus fronting, i.e. the identity of Y in the tree in (37).

3.2.4 Establishing the landing site

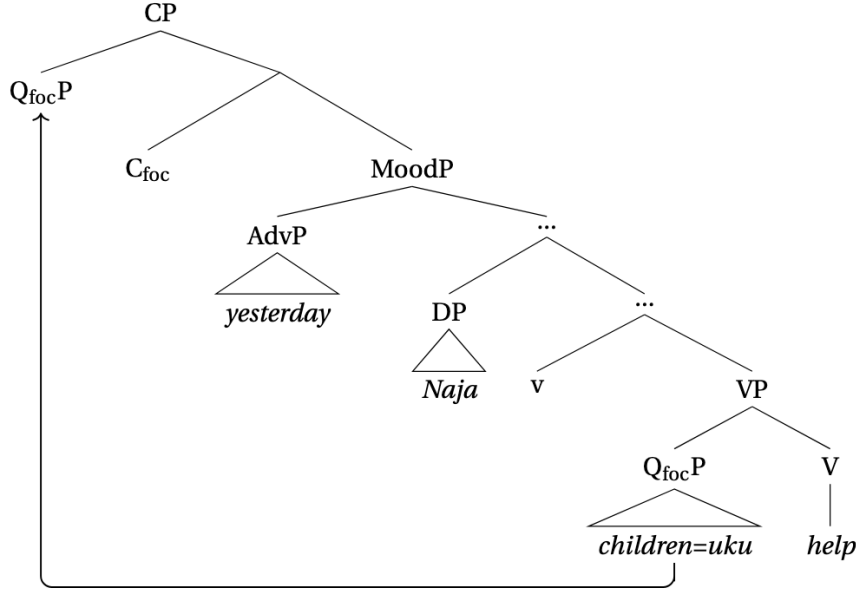
An important property of the focus fronting construction is that the focused constituent **MUST** be initial.

- (45) **Meeqqa-n=uku** ippassaq Naja-p ikior-ai.
child-PL.ABS=FOC.PL yesterday Naja-ERG help-3SG>3PL.PART
It was the children Naja helped yesterday. Third author CEG-GS-2
- (46) *Ippassaq **meeqqa-n=uku** Naja-p ikior-ai.
yesterday child-PL.ABS=FOC.PL Naja-ERG help-3SG>3PL.PART
Intended: It was the children Naja helped yesterday. Third author CEG-GS-3
- (47) *Ippassaq Naja-p **meeqqa-n=uku** ikior-ai.
yesterday Naja-ERG child-PL.ABS=FOC.PL help-3SG>3PL.PART
Intended: It was the children Naja helped yesterday. Third author CEG-GS-4

¹²The last example is from a text curated by Maria Bittner (<https://sites.google.com/view/maria-bittner/kalaallisut>). We have reglossed it in accord with the glossing conventions employed here and also changed the spelling to match the current standard orthography.

In the grammatical (45), the focused object, *meeqqan-*, has moved across the subject *Naja-p* and across the adverb *ippassaq*, placing it in initial position. Moving it to a lower position, as in (46), or not at all (47) leads to ungrammaticality. We therefore analyze focus fronting as obligatory movement to specifier of CP as illustrated in (48). (The mechanics of this movement are laid out in section 4.4.)

(48) FOCUS FRONTING IS MOVEMENT OF $Q_{foc}P$ TO SPECIFIER OF CP



The landing site of focus movement is thus higher than the landing site of hyperraising, specifier of v , and yet only the latter can cross a finite clause boundary. This is unexpected in light of the Height Locality Connection as schematized in the tables below.

(49) EXPECTED CORRELATION OF HEIGHT AND LOCALITY

landing site	cross-clausal
high	✓
low	✗

(50) CORRELATION OF HEIGHT AND LOCALITY IN KALAALLISUT

landing site	cross-clausal
high	✗(focus movement)
low	✓ (hyperraising)

Having established the locality profiles of hyperraising and focus fronting, we now turn to developing an analysis that accounts for this difference.

4 A phase-unlocking analysis

Up til now we have framed the difference between hyperraising and focus fronting in terms of their respective landing sites, i.e. the location of the probe that drives the movement. In hyperraising the probe is on v , in focus fronting it is on C . We will now offer an analysis in which the key difference between hyperraising and focus fronting is not the location of the probe (v vs. C), but the specification of the probe. Specifically,

hyperraising involves conjunctive probing for ϕ and D, whereas focus fronting involves non-conjunctive probing for focus with no requirement that the goal be a certain syntactic category. In other words, the hyperraising probe is “picky” and the focus probe is not. Because the embedded CP can ϕ -Agree with the hyperraising probe without satisfying it, that Agree relation UNLOCKS the embedded CP for further ϕ -probing, and thereby enables hyperraising. The focus probe on C, however, cannot unlock the embedded CP for extraction and focus fronting from within and the embedded CP is therefore impossible.

4.1 Agree

I adopt the Interaction-Satisfaction theory of Agree presented in Deal (2015, 2022, to appear). In this theory a probe is specified for three things: i) what features it can INTERACT with, ii) what features SATISFY it and make it halt and iii) whether or not it triggers movement. These specifications are represented in the notation in (51), where F = feature, INT = interaction, SAT = satisfaction, and M = move:

$$(51) \quad [\text{INT:F}_1, \text{SAT:F}_2^M]$$

This probe can interact with any accessible goal bearing the feature F_1 . It is satisfied by feature F_2 , which may be identical to F_1 (e.g. the ϕ -probe on English T interacts with ϕ and is satisfied by ϕ) or may be distinct (e.g. the topicalization probe in Mandarin interacts with topic and is satisfied by D; Chou 2025). Finally, the superscripted M indicates that the probe moves the goal that satisfies it to its specifier. As noted earlier, conjunctive probing plays a key role in the analysis of hyperraising. In conjunctive probing, two (or more) features need to be co-present on the goal for the probe to successfully Agree with it.¹³ A conjunctive probe is therefore “picky”, requiring the co-presence of features on the goal, whereas a non-conjunctive probe is less “picky”, requiring only a single feature on the goal. For instance, in Ndengleko the focus probe is only satisfied by a goal that is both focus and nominal (Scott 2021). In Dinka the A-bar probe responsible for relativization, topicalization and focus movement probe simultaneously probes for ϕ so any potential goal lacking ϕ -features is bypassed (Van Urk 2015). Similarly Phan and Chou (2025) argue that Vietnamese locative inversion constructions limit A-bar probing to goals bearing a D feature. Finally, in Nukouro, the relativization probe is only satisfied by an absolutive goal (Drummond 2023).¹⁴ We notate conjunctive probing with the plus sign:

$$(52) \quad [\text{INT:F}_1, \text{SAT:F}_2+\text{F}_3^M]$$

The probe in (52) can interact with any goal bearing F_1 , but is only satisfied by a goal bearing F_2 and F_3 . As is the case for non-conjunctive probing, F_1 and F_2 may be distinct or identical. F_3 is distinct from F_2 .

4.2 Phase unlocking

Phase unlocking has been applied to a range of phenomena, including i) cross-clausal *wh*-extraction in Tagalog (Rackowski and Richards 2005) and Dinka (van Urk and Richards 2015), ii) extraction from DP in Northern Ostyak and Blackfoot (Branan 2018), iii) extraction from Ch’ol comitatives (Little 2025), iv) hyperraising to subject in Cantonese (Lee and Yip 2024) and Vietnamese (Lee and Yip 2024) and v) cross-clausal passivization in Northern Tundra (Franco Liu p.c.). Here we adapt the mechanism to explain why hyperraising can cross a finite clause boundary in Kalaallisut, but focus fronting cannot.

¹³ Conjunctive probing is a subtype of composite probing along side disjunctive and independent probing. Only conjunctive probing is relevant here. See Scott (2021) and Lohninger et al. (2022: 35-41) for further discussion and references.

¹⁴ A precedent for conjunctive probing is Coon and Bale’s (2014) notion of ‘fused probes’, which they utilize to account for the interaction between person and number agreement in Mi’gmaq.

The intuition behind phase unlocking is fairly simple: phases block off material inside them for further syntactic operations, specifically for Agree. However, if a probe outside the phase can Agree with the phase itself, that unlocks the phase and the probe can continue its search into the phase and attract a phrase from within the phase. A detailed derivation of phase-unlocking is given in the next section.

4.3 Phase unlocking in hyperraising

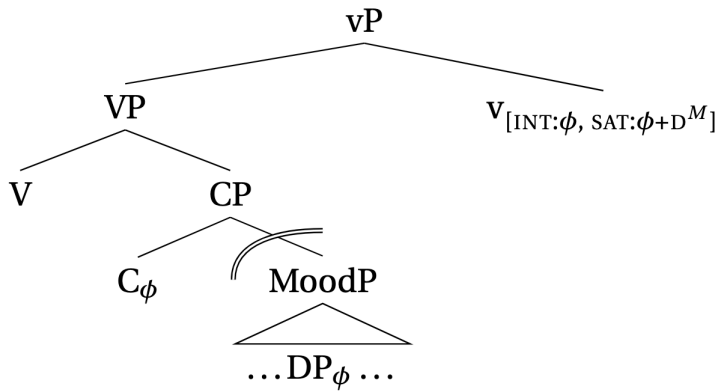
The analysis of phase unlocking in Kalaallisut hyperraising to object builds on Halpert’s (2019) analysis of hyperraising raising to subject in Zulu. The core idea is that the embedded CP can INTERACT with the raising probe but does not SATISFY it. This, together with phase unlocking, allows the hyperraising probe to probe into the embedded CP, interact with the embedded subject, and move it out of the embedded clause to its specifier. This is achieved by the probe specification in (53).

$$(53) \quad [\text{INT}:\phi, \text{SAT}:\phi + \text{D}^M]$$

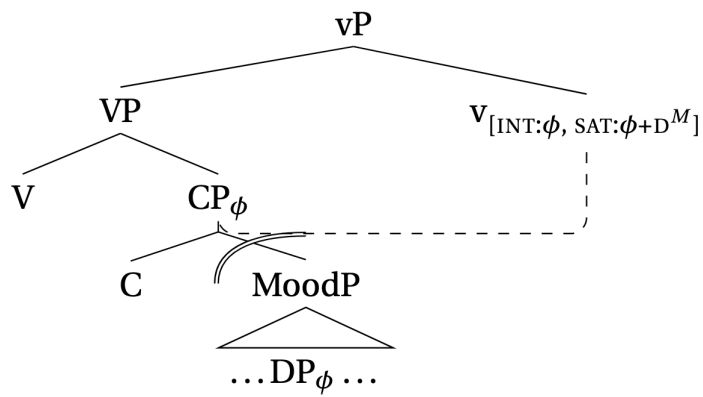
Recall that CPs control object agreement on transitive embedding verbs, indicating CPs bear ϕ -features in Kalaallisut. The probe in (53) will therefore interact with the embedded CP—and thereby unlock it—but not be satisfied by it, since CP doesn’t bear a D feature. Hence probing continues until the probe finds the subject DP and raises it to its specifier. In Zulu this probe is on T and derives raising to subject position. In Kalaallisut it is on v and derives raising to a lower object position.

Derivation of hyperraising Below is a step-by-step derivation of hyperraising. The double arch indicates that the sister of the C phase head is sealed off for Agree with a probe above the phase. This leaves the edge accessible to operations with probes outside the phase, though that will not play a role until section 8.

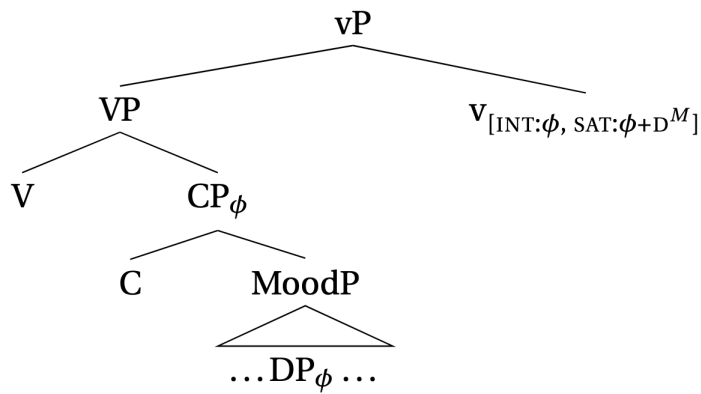
1. Embedded CP Merges with matrix V and locks
2. v with its hyperraising probe is Merged



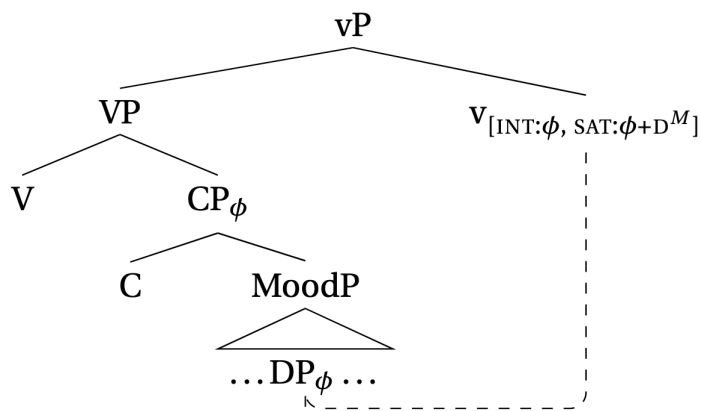
3. Hyperraising probe interacts with CP and copies its 3SG features



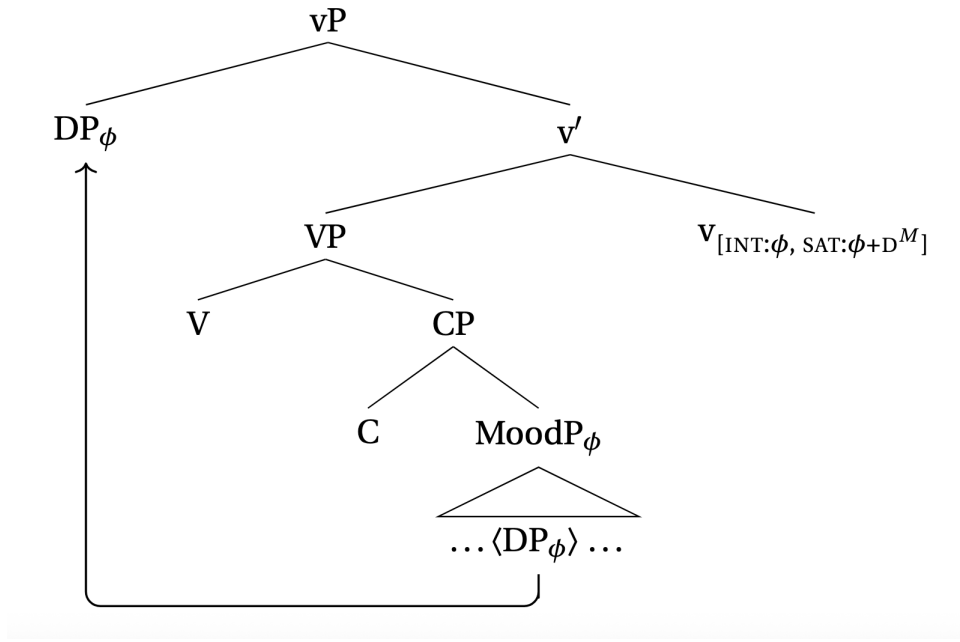
4. As a result CP phase is unlocked



5. Hyperraising probe is not satisfied and probes into CP



6. Hyperraising probe copies features from subject DP, moves that DP to its specifier and halts



4.4 No phase unlocking in focus fronting

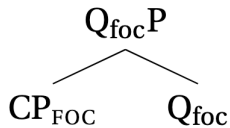
In contrast to the hyperraising probe, the focus probe on matrix C is not picky. It will interact with the first $Q_{\text{foc}}P$ it encounters, be satisfied, move the $Q_{\text{foc}}P$ to its specifier and halt. This behavior is captured with the probe specification in (54).

$$(54) \quad [\text{INT}:Q_{\text{foc}}, \text{SAT}:Q_{\text{foc}}^M].$$

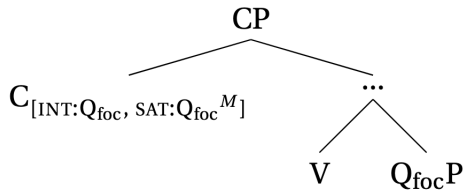
With this probe specification in hand, we give two key derivations. One in which the embedded CP is focused and nothing else is and a second derivation in which the only element bearing focus is inside the embedded CP, mirroring the configuration in hyperraising. These derivations are given in a step-wise fashion below.

Derivation 1: embedded CP is focused and nothing else is

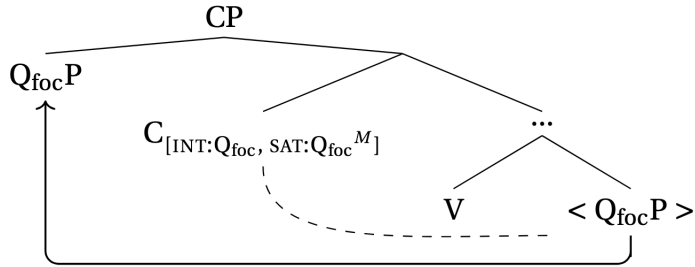
1. Embedded CP_{foc} is assembled and locked
2. CP Merges with Q_{foc}



3. Matrix V Merges with $Q_{\text{foc}}P$
4. Matrix MoodP is assembled
5. Matrix C is Merged with its focus probe ($[\text{INT}:Q_{\text{foc}}, \text{SAT}:Q_{\text{foc}}^M]$)



6. Focus probe probes $Q_{foc}P$ and Moves it to the specifier of matrix C



7. Probe is satisfied and halts

This derivation yields the grammatical string in (55).

- (55) **Angerla-jaar-tutin=una** uanga eqqaama-giga.
 come.home-early-2SG.PART=FOC 1S.ERG remember-1SG>3SG.PART

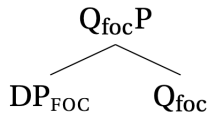
It was that you came home early that I remember.

Third author CEH-GS-5

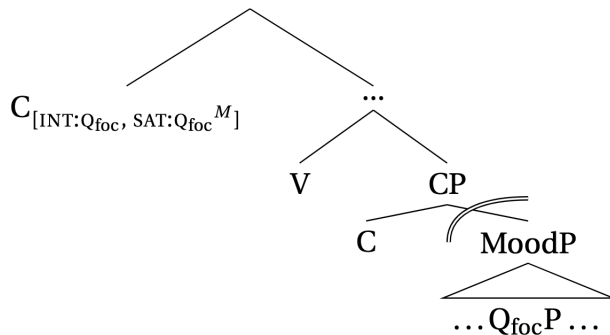
The second derivation mimics a hyperraising configuration: the desired goal is inside the embedded CP and the probe is outside the embedded CP.

Derivation 2: CP does not bear focus but embedded DP does

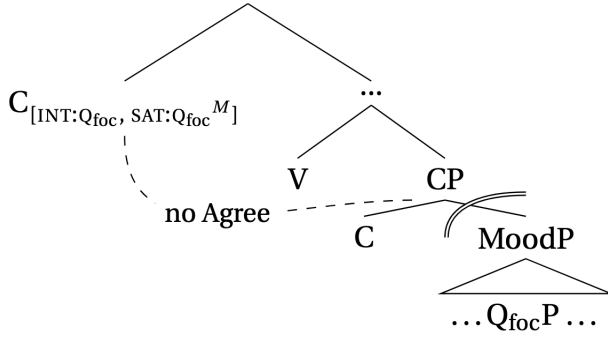
1. Embedded focused DP Merges with Q_{foc} .



2. Embedded CP is assembled and locked.
 3. Matrix V is Merged with embedded CP.
 4. Matrix MoodP is assembled.
 5. Matrix C is Merged with its focus probe.

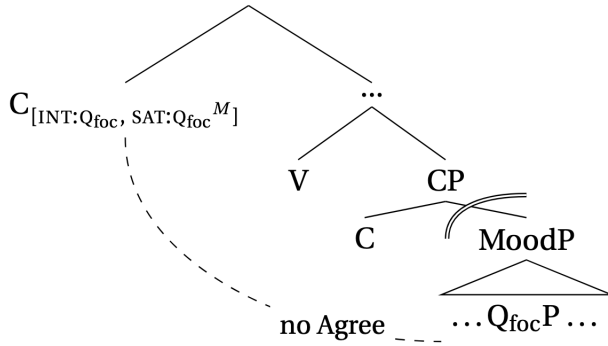


6. Focus probe attempts to Agree with embedded CP but cannot interact with it because the embedded CP is not encased in a Q_{foc} shell.



7. Embedded CP remains locked.

8. Focus probe cannot probe past the locked CP to Agree with $Q_{foc}P$.



9. Since the probe has exhausted its search space, it halts and nothing moves.

Derivation 2 demonstrates that cross-clausal focus movement cannot be derived, which accounts for the ungrammaticality of (56) and (57).¹⁵

- (56) ***Meeqqa-n=uku** sinnattor-tunga [Naja-p ___ ikior-ai].
 child-PL.ABS=FOC.PL dream-1SG.PART Naja-ERG help-3SG>3PL.PART
 Intended: It's the children that I dreamt Naja helped. Second author CAU-4
- (57) ***Meeqqa-n=uku** ilimagi-giga [illit ___ ikior-ikit].
 child-PL.ABS=FOC.PL expect-1SG>3SG.PART 2SG.ERG help-2SG>3PL.PART
 Intended: It is the children that I expect you to help. Second author CCL-1

¹⁵ Assuming that Agree can fail (Preminger 2009) the derivation above would generate the ungrammatical string in (i):

- (i) *Sinnattor-tunga [Naja-p **meeqqa-n=uku** ikior-ai].
 dream-1SG.PART Naja-ERG child-PL.ABS=FOC.PL help-3SG>3PL.PART
 Intended: It's the children that I dreamt Naja helped. CGU-GS-1

The question thus arises: what rules out (i)? Here we tentatively suggest an answer in terms of GOAL FLAGGING (Deal to appear, Clem and Deal 2024). When C_{foc} Agrees with $C_{foc}P$, it deposits its C_{foc} feature on Q_{foc} and that feature is in turn referenced by the Vocabulary Items for =una and =uku. So only if the C_{foc} feature is present on Q_{foc} , can the focus clitic be inserted. In Derivation 2 there is no Agree between C_{foc} and $Q_{foc}P$, because the embedded CP phase that contains $Q_{foc}P$ stays locked. Consequently, the C_{foc} feature is not present on Q_{foc} at Vocabulary Insertion and hence =uku cannot be inserted. Thus, there is no way to derive (i) either.

The basic pattern is now accounted for. Hyperraising is possible because the ϕ -probe on matrix *v* unlocks the embedded CP for movement from within. Cross-clausal focus fronting is impossible because the focus probe on matrix *C* does not unlock the embedded CP for movement from within. This logic can be generalized as two conditions that must both be met for phase unlocking to take place (cf. Halpert 2019:143):

1. CONDITION I: the relevant probe can INTERACT with the embedded CP
2. CONDITION II: the relevant probe **is not** SATISFIED by the embedded CP

Hyperraising satisfies both conditions, but focus fronting can only ever satisfy one. If the embedded CP is focused it can interact with the probe (via Q_{foc}), but it also satisfies it in violation of Condition II. That is derivation 1 above. Conversely, in derivation 2, the focus probe cannot interact with the embedded CP in violation of Condition I.

Number inflection The realization of Q_{foc} as *=una* or *=uku* is governed by the number feature of the focused DP. If that DP is plural, Q_{foc} is realized as *=uku* (58); if it is singular, Q_{foc} is realized as *=una* (59).

(58) meeqqa-n=*uku*
child-PL=FOC

(59) meerar=*una*
child=FOC

So far we have been silent on how this difference is derived, but it is important for understanding the interaction between hyperraising and focus fronting discussed in section 5.1, so we will tackle it here. Our proposal is that the differing number morphology on the Q_{foc} head arises from Q_{foc} probing its sister and copying back its ϕ -features, which include number. More precisely, this probe is specified to interact with any feature (F) and be satisfied by focus: [INT:F, SAT:FOC]. When it probes the sister of Q_{foc} , i.e. the focused phrase, it copies all the features of that goal, including focus, category feature and ϕ , if present. It is satisfied by the focus feature and halts. At Vocabulary Insertion, the copied features govern the insertion of *=una* vs. *=uku*. The plural form *=uku* is inserted if it copied plural, whereas *=una* is inserted if it didn't copy plural.

(60) [FOC PL] $\leftrightarrow$ *=uku*

(61) [FOC] $\leftrightarrow$ *=una*

There are thus two contexts where *=una* will be inserted: when the probe on Q_{foc} copied singular (as in (59)) and when there was no number feature to copy.¹⁶ This latter situation is found where the focused phrase is a ϕ -less adverbial as in (62) and (63). In these ϕ -less contexts, the focus particle must surface as *=una*, indicating that *=una* is the elsewhere form:

(62) **Ippassar=*una*** meeqqa-t Naja ikior-aat.
yesterday=FOC child-PL.ERG Naja.ABS help-3PL>3SG.PART

It was yesterday that the children helped Naja.

Second author AOV-1

¹⁶If one assumes that there is no singular feature in the grammar of Kalaallisut, only plural, then these two conditions collapse into one. The correct forms would still be derived with the Vocabulary Items in (60) and (61).

- (63) **Qanor=una** oqar-pit?
 how=FOC say-2SG.INT

What is it that you are saying? [lit. How is it that you are saying?]

Second author BIN-2

In (62) the probe on Q_{foc} copies Adv and focus from *ippassaq*. The most specified Vocabulary Item matching this feature combination is (61) and *=una* is inserted. In (63) the Q_{foc} probe copies adv, wh and focus and, again, the most specified Vocabulary Item that matches this feature combination is (61) and the elsewhere form *=una* is inserted.

5 Two predictions

The analysis developed above makes two predictions. First, hyperraising should feed focus fronting and any movement driven by features that cannot unlock the CP phase should be clause bounded. As we show in turn, both are borne out.

5.1 Hyperraising feeds focus fronting

First, recall that on its own focus-fronting cannot cross a finite clause boundary:

- (64) ***Illin=una** uanga eqqaama-giga angerla-jaar-tutit.
 2SG.ABS=FOC 1SG.ERG remember-1SG>3SG.PART come.home-early-2SG.PART

Intended: It is you that I remember came home early.

Second author CCG-1

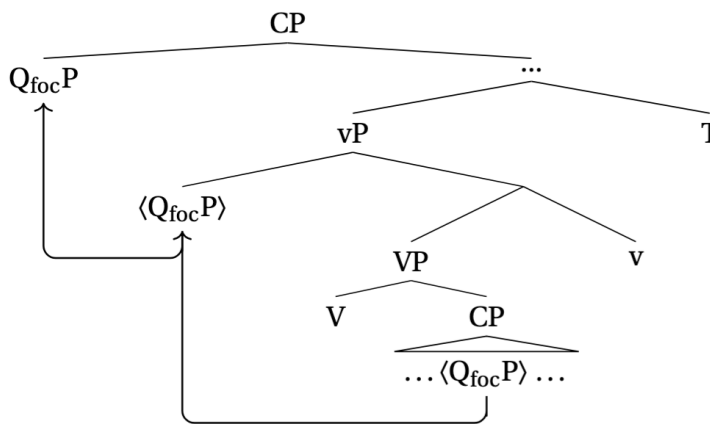
However, if hyperraising brings a focused DP into the matrix clause, we predict that the DP can undergo subsequent focus fronting, because that focus movement would be within a single clause. This is indeed the case, as seen in (65). The schematic tree in (66) illustrates the two-step movement.

- (65) **Illin=una** uanga eqqaama-gikkit angerla-jaar-tutit.
 [2SG].ABS=FOC 1SG.ERG remember-1SG>[2SG].PART come.home-early-2SG.PART

It is you that I remember came home early.

Second author CCG-3

(66)



How does the first movement step come about, given that the focused DP is encased in a $Q_{\text{foc}}P$? Recall from the discussion of number inflection that the probe on Q_{foc} is specified to interact with any feature ([INT:F, SAT:FOC^M]) and therefore copies all features of the goal. If the focused element is a DP, $Q_{\text{foc}}P$

ends up with at least three features: ϕ , D, and focus. The first two allow it to satisfy the hyperraising probe ([INT: ϕ SAT: $\phi+D^M$]) and the latter the focus probe on matrix C. Since both are movement probes, $Q_{\text{foc}}P$ moves twice as in (66).

The grammaticality of (65) also shows that the ungrammaticality of (64) is not due to a semantic or pragmatic problem with the intended content. The problem is syntactic: focus fronting cannot cross a clause boundary.

5.2 *Wh*-movement

According to the phase unlocking analysis developed above, cross-clausal movement is only possible when two conditions are met:

1. CONDITION I: the relevant probe can INTERACT with the embedded CP
2. CONDITION II: the relevant probe **is not** SATISFIED by the embedded CP

We have already seen that focus fronting fails either Condition I (derivation 2) or Condition II (derivation 1) and is therefore clause bounded. Here we show that *wh*-movement is also correctly predicted to be clause bounded.

Following Cable (2010: 38ff), the *wh*-phrase is encased in a Q_qP (the subscripted *q* stands for ‘question’ and is there to distinguish this particular *Q* head from others, including Q_{foc}). An interrogative *C* head probes Q_qP and moves it to specifier of that CP. The probe in question is specified [INT: Q_q , SAT: Q_q^M]. If Q_qP is inside an embedded CP, Q_qP is inaccessible to the matrix probe: there is no phase unlocking because the probe cannot interact with the non-interrogative embedded CP and hence cannot unlock it (failing Condition I).

It is thus impossible to derive cross-clausal *wh*-movement, and it is indeed ungrammatical:

- (67) *Kikkut sinnattor-pit Naja-p ikior-ai?
 who.PL.ABS dream-2SG.INT Naja-ERG help-3SG>3PL.PART
 Who all did you dream that Naja helped? Second author CAU-2
- (68) *Kikkut ilimagi-viuk Naja ikior-aat?
 who.PL.ERG expect-2SG>3SG.INT Naja.ABS help-3PL_i 3SG.PART
 Intended: Who all did you expect to help Naja? Third author CEQ-GS-1

As with focus fronting, hyperraising can feed *wh*-movement in the matrix clause giving the appearance of cross-clausal *wh*-movement in (69).

- (69) Kikkut ilimagi-vigit Naja ikior-aat?
 who.PL.ERG expect-2SG>3PL.INT Naja.ABS help-3PL>3SG.PART
 Who all did you expect to help Naja? Third author CEQ-GS-2

Summing up, the phase unlocking analysis captures the basic locality difference between hyperraising and focus fronting. It accounts for apparent cross-clausal focus movement: it is in fact within-clause focus fronting fed by hyperraising. Finally, it extends to capture the clause-boundedness of *wh*-movement.

- (70) LOCALITY PROFILES OF KALAALLISUT MOVEMENT OPERATIONS

landing site	cross clausal
high	✗focus fronting
high	✗ <i>wh</i> -movement
low	✓hyperraising

6 Alternative analyses

Here we consider two alternative analyses of the locality contrast between Kalaallisut hyperraising and focus fronting: Delayed Opacity (Deal 2017) and Horizons (Keine (2020)). While both are successful in capturing the Kalaallisut data considered so far, we will argue that the phase unlocking analysis makes better predictions than the Delayed Opacity analysis and is simpler and more explanatory than the Horizon analysis.

6.1 Delayed Opacity

Deal (2017) investigates hyperraising to object in Nez Perce (Sahaptian, Idaho) and develops an analysis in terms of DELAYED OPACITY. The intuition behind the Delayed Opacity analysis is that a low probe can “pull” a phrase out of a phase before that phase seals off, but a high probe cannot. In other words, contra Keine’s Heigh-Locality Connection, lower is freer. Crucially LOW means below then next up phase head and HIGH means above the next up phase head. Specifically, if a probe is below a matrix phase head, it can interact with a goal inside an embedded clause, because the embedded CP phase is not sealed off until the matrix phase head is Merged. In contrast, if a probe is above the matrix phase head it cannot probe into the embedded CP, because the matrix phase head seals it off. To understand the predictions of the two analyses, we need to establish which categories are phase heads in Kalaallisut. Here we rely on Compton and Pittman’s (2010) argument that vP is not a phase in Inuit, and that the only phase heads are C and D. Under this assumption, the Delayed Opacity analysis accounts equally well for the data considered so far: because the hyperraising probe is both low (below the matrix C phase head) **and** has the ability to unlock the CP phase via its ϕ -features, both analyses predict hyperraising to be possible. Because the focus fronting probe is both high (not below the matrix C phase head) **and** lacks the ability to unlock the CP phase, both analyses predict cross-clausal focus fronting to be impossible.¹⁷

(71) SHARED PREDICTIONS OF DELAYED OPACITY AND PHASE UNLOCKING

	unlocking feature & low probe	non-unlocking feature & high probe
Delayed Opacity	✓	✗
Phase unlocking	✓	✗

In principle, however, it is possible to tell the two analyses apart. If a probe with an unlocking feature is high, the unlocking analysis predicts cross-clausal movement to be possible, whereas the Delayed Opacity analysis predicts it to be impossible. Conversely, if the probe is non-unlocking and low, the phase unlocking analysis predicts cross-clausal movement to be impossible, whereas the Delayed Opacity analysis predicts it to be possible. These predictions are schematized below.

(72) DIFFERING PREDICTIONS OF DELAYED OPACITY AND PHASE UNLOCKING

	unlocking feature & high probe	non-unlocking feature & low probe
Delayed Opacity	✗	✓
Phase unlocking	✓	✗

I am not aware of any phenomena in Kalaallisut that instantiates the middle column. It is true that regular clausal agreement has been argued to involve a high probe—in C by Compton (2018) and in dedicated Agr projections by Yuan (2018)—but if v is not a phase, Agree with Voice-internal subject and object DP goals

¹⁷Deal (2017:12) makes it clear that to be able to probe into a phase the probe must be *below* the next phase head up. Since the FOC probe is *on* the C phase head and not below it, the Delayed Opacity analysis correctly rules out cross-clausal focus fronting.

will involve regular within-phase Agree under either analysis. I therefore set the middle column aside and turn to the rightmost column: are there instances of a non-unlocking feature on a low probe? Here we suggest that this is instantiated by low *wh*-movement. The examples of *wh*-questions considered so far all have the *wh*-phrase in initial position, which we assume to be specifier of CP. However, Kalaallisut also have *wh*-medial questions:

- (73) Meeqqa-t kina ikior-paat?
 child-PL.ERG who.ABS help-3PL>3SG.INT
 Who has the children helped?

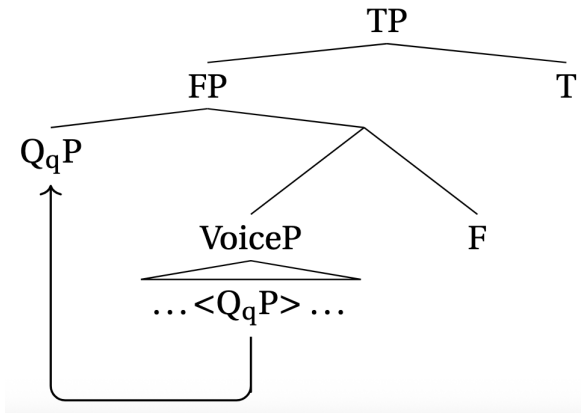
Second author AGO-2

- (74) Illit kia-p ikior-paatit?
 2SG.ABS who-ERG help-3SG>2SG.INT
 Who helped you?

Third author CCJ-GS-2

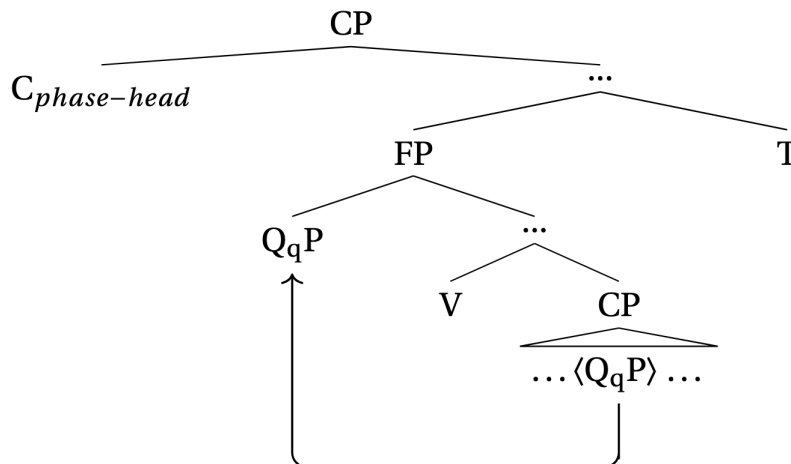
Following Sherkina-Lieber (2005) analysis of Inuktitut, we assume that *wh*-medial orders involve short *wh*-movement to a position between TP and VoiceP. As above, *wh*-movement is implemented as Q_qP movement.

(75)



With these two assumptions in place, we can consider the predictions made by either analysis. The phase unlocking predicts low *wh*-movement into the matrix clause to be ungrammatical, because *wh* is not an unlocking feature. The Delayed Opacity analysis, in contrast, predicts low *wh*-movement into the matrix clause to be possible because it is to a position (spec-FP) below the next up phase head (matrix C):

(76)



In fact cross-clausal low *wh*-movement is ungrammatical:

- (77) *Juuna kikkut oqar-pa angerla-jaar-ner-sut?
 Juuna.ABS who.PL.ABS say-3SG.INT come.home-early-if-3PL.PART
 Intended: Who all did Juuna say came home early? Third author BUQ-GS-2
- (78) *Naja-p suna oqarfigi-vaatit meeqa-t tillik-kaat?
 Naja-ERG what.ABS say.to-3SG>2SG.INT child-PL.ERG steal-3PL>3SG.PART
 Intended: What did Naja tell you that the children stole? Third author CDX-GS-4

If the analysis of low-wh movement is corroborated, the ungrammaticality of (77) and (78) would provide an argument in favor of the unlocking analysis over the delayed opacity analysis.

6.2 Horizons

In addition to arguing for the Height-Locality Connection, Keine (2020) develops a theory of locality, according to which individual probes may be associated with a HORIZON.¹⁸ The horizon is a particular category feature that determines which domains are inaccessible to the probe. For instance, if a probe has horizon T it cannot probe into TP. If a probe lacks a horizon, it can keep probing. In this system, the Kalaallisut hyperraising probe lacks a horizon, whereas the Q_{foc} probe on C has horizon C. The latter means that the focus probe will abort its search when it encounters the embedded CP and therefore be unable to probe into it. This correctly captures the clause-boundedness of focus fronting. Setting phases aside for the moment, leaving the hyperraising probe without a horizon also seems to capture its behavior: nothing forces it to abort its search upon encountering CP and it can therefore probe into the CP and hyperraise an embedded DP. However, Keine (2020) argues that there are two types of locality constraints: horizons and phases. If so, the question of how the hyperraising probe probes into the embedded CP—given that it is a phase—is left unanswered. We could add phase unlocking to the horizon system and create a double-tiered analysis that employs both phase unlocking and horizons. If the phase unlocking analysis can capture the same facts on its own, that seems preferable to a dual analysis. A second point is that the phase-unlocking analysis is grounded in language specific facts that the horizon analysis does not take into account. First, the ability of the hyperraising probe to unlock an embedded CP is predicated on the fact that CPs bear ϕ -features in the language. That connection is lost under the horizon analysis, where the larger search domain of the hyperraising probe is attributed to its lack of a horizon. Second, the fact that focus fronting and *wh*-movement do not unlock the embedded CP is a consequence of the associated probes not being picky. That in turn is forced by distributional facts: as shown in section 3.2, any phrase capable of being focused can focus front. Similarly, there is no category restriction on which phrases can front in *wh*-questions, indicating a non-picky probe. On the horizon analysis, the difference between cross-clausal hyperraising on the one hand and clause-bounded focus fronting and *wh*-movement on the other is simply an accident.

7 Generalized vs. relativized phase unlocking

Considering phase unlocking as a mechanism, a question arises: does unlocking by a particular probe leave the phase open for other probing? We can sharpen this question in terms of two conceptions of phase unlocking.

¹⁸I thank Stefan Keine for discussion of a horizon analysis of the Kalaallisut facts.

Generalized phase unlocking Agree with a phase for feature F unlocks the phase for any subsequent probing.

Hyperraising of XP → cross-clausal focus fronting/*wh*-movement of YP

Relativized phase unlocking Agree with a phase for feature F only unlocks the phase for probing for F¹⁹

Hyperraising of XP ⇏ cross-clausal focus fronting/*wh*-movement of YP

The examples below show that hyperraising does not enable focus fronting or *wh*-movement out of the embedded CP suggesting that phase unlocking is relativized in Kalaallisut. Consider first the focus fronting example in (79):

- (79) ***Meeqqa-n=uku** illit eqqaama-gikkit ikior-itit.
 child-PL.ABS=FOC.PL [2SG].ABS remember-1SG>[2SG].PART help-2SG>3PL.PART
 Intended: It is the children I remember you helped. Second author CBZ-1

This derivation combines hyperraising of the subject (boxed) and focus fronting of the object (bolded). The ungrammaticality of (79) shows that ϕ -Agree between the hyperraising probe and the embedded CP phase only opens the gate for further ϕ -probing by the hyperraising probe, and not for later probing by the focus probe in C. The alternative offered for (79) reverts to a feeding relationship between hyperraising and focus fronting: the subject ‘you’ first hyperraises and then focus fronts:

- (80) **Illin=una** eqqaama-gikkit meeqqa-t ikior-itit.
 [2SG].FOC remember-1SG>[2SG] child-PL.ABS help-2SG>3PL.PART
 Intended: It is you I remember helped the children. Second author CBZ-2

If phase unlocking is indeed relativized, we expect the same pattern for *wh*-extraction and that is what we find. Hyperraising the subject also does not enable *wh*-movement of the object ‘who’ out of the embedded CP (81), whereas a feeding construction is possible (82).

- (81) *Kina meeqqa-t eqqaama-vigit ikior-aat?
 who.ABS child-[PL].ABS remember-2SG>3[PL].INT help-3PL>3SG.PART
 Intended: Who do you remember that the children helped? Third author CCF-GS-6
- (82) Kikkut eqqaama-vigit angerla-jaar-tut?
 who.[PL].ABS expect-2SG>3[PL].INT come.home-early-3PL.PART
 Who all do you remember came home early? Second author BZB-2

Summing up Kalaallisut phase unlocking we can say that a phase is not unlocked for all. It is only unlocked for the feature that unlocked it. Therefore hyperraising does not give a free ride to independent focus fronting or *wh*-movement. As far as we can tell the distinction between generalized and relativized phase unlocking has not received much attention, though Lee and Yip (2024) touch on it in a footnote in connection with their analysis of hyperraising to subject in Vietnamese and Cantonese. They write “The current proposal does not predict that the CP is transparent for all syntactic operations. It is just “unlocked” for the Probe of the first Agree relation ...—it is relativized to the respective head which agrees with the phasal CP.” Turning to other instances of phase unlocking from the literature a different picture emerges: unlocking by X facilitates probing by Y. For instance, in their analysis of Tagalog extraction, Rackowski

¹⁹It would also be possible to relativize phase-unlocking to probes, and not features. The data at hand is compatible with relativizing to a feature or to a probe and I leave it open whether both exist.

and Richards (2005) assume that agreement with an embedded CP phase for case opens it up for *wh*-movement and van Urk and Richards (2015) essentially makes the same assumption for Dinka A-bar extraction more generally. Branan (2018) assumes for a range of languages that ϕ -Agree with a DP opens it up for *wh*-subextraction from within and Little (2025) argues that ϕ -Agree with a C’hol comitative DP opens it up for *wh*-subextraction. It thus appears that both generalized and relativized phase unlocking are attested, which raises interesting theoretical questions. Does the generalized vs. relativized nature of phase unlocking depend on the features involved? If not, is it a parameter of sorts? If it is, is it set by language or by construction. More work is needed to gain traction on these questions.

8 Conclusion and cross-linguistic outlook

The main conclusion of this paper is that Keine’s Height-Locality Connection cannot be maintained as a universal: lower (hyperraising) is sometimes freer than higher (focus fronting). The lack of cross-clausal focus fronting and *wh*-movement in Kalaallisut is not determined by the **height** of the probe that triggers the movement, but by the **feature** that is being probed for.

Once we look more broadly, we find that all combinations of Height Locality connections are attested^{20,21}

- (83) HIGH LESS RESTRICTED THAN LOW (Atchan, English, Nuntajijji, Tiwa, (a variety of) West Circassian, ...)

landing site	cross-clausal
high	✓
low	✗

- (84) LOW LESS RESTRICTED THAN HIGH (Kalaallisut)

landing site	cross-clausal
high	✗
low	✓

- (85) HIGH AND LOW EQUALLY RESTRICTED (Polish, Russian, Tsez)

landing site	cross-clausal
high	✗
low	✗

²⁰✗ and ✓ represent specific movement operations, not the language as a whole. For instance, the ✓ in the table above indicates that there is a least one movement operation to a high position that can cross a finite clause boundary and ✗ indicates that there is at least one movement operation to a low position that cannot cross a finite clause boundary.

²¹I am grateful to the speakers and linguists who have contributed data and insight into this larger cross-linguistic project: Shweta Akolkar (Hindi), Kenneth Baclawski (Eastern Cham), Tanya Bondarenko (Georgian), Maddy Bossi (Kipsigis), Max Dabkowski (A’ingae), Ginny Dawson (Tiwa), Alex Elias (Jao), Ksenia Ershova (West Circassian), Becky Everson (Shekgalagadi), Steven Foley (Georgian), Vera Gribanova (Russian), Jorge Hankamer (Turkish), Larry Hyman (Limba), Becky Jarvis (Atchan), Tyler Lemon (Uab Meto, Vietnamese), Franco Liu (Northern Tadjik), Wendy López-Marquez (Nuntajijji), Lev Michael (Nihagantsi languages), Léa Nash (Georgian), Zach O’Hagan (Caquinte), Wesley dos Santos (Brazilian Portuguese, Kawahiwa), and Hannah Sande (Guebié).

(86) HIGH AND LOW EQUALLY UNRESTRICTED (Brazilian Portuguese, Kispigis, ...)

landing site	cross-clausal
high	✓
low	✓

On Keine's (2020) formulation of the Height Locality Connection in (87), only the pattern in (83) should be allowed.

(87) HEIGHT-LOCALITY CONNECTION Keine (2020:12)

Movement types differ in their landing sites. The higher the landing site of a movement type is in the clausal structure, the more kinds of structures are transparent to this movement type.

Patterns of equal (un)restrictedness violate (87) as does the 'low less restricted than high' pattern found in Kalaallisut. One could consider the weaker version of Keine's Height Locality Connection in (88).

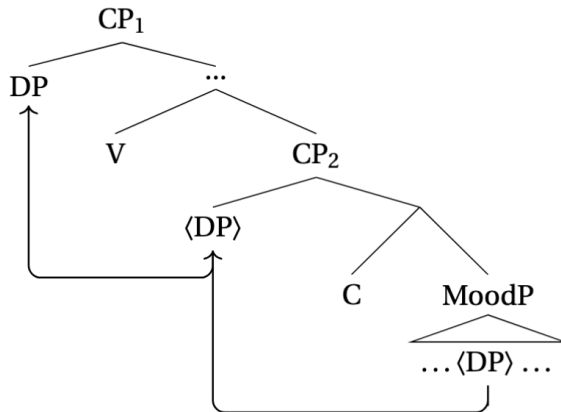
(88) WEAK HEIGHT-LOCALITY CONNECTION

Movement to a higher landing site is at least as restricted as movement to a lower landing site.

The formulation in (88) rules in two additional patterns: equally restricted movement to higher and lower landing sites (85) and equally unrestricted movement to higher and lower landing sites (86). Under either formulation, Kalaallisut constitutes a counter example, and the only one I know of. As such, the present study makes a typological contribution by demonstrating that a pattern ruled out by current assumptions is in fact attested. It also raises the question of whether the Kalaallisut pattern is unique or whether there are more languages with this profile.

A second relevant question is whether the proposed analysis of Kalaallisut shines any light on the availability of all four patterns. It might be the beginning of an explanation, but we need a second point of variation in addition to whether there is phase-unlocking or not. Here we appeal to the assumption that the edge of a phase is transparent to operations from the outside. There are thus two ways for a phrase to move across a finite clause boundary, i.e. across a CP phase:

1. the phase can be unlocked, by Agree, making the phrase free to move to a higher clause OR
2. the phrase can move to the edge of the embedded phase and from there more into the higher clause (INTERMEDIATE WH-MOVEMENT)



These two parameters yield a typology that captures the observed patterns:

(89)		PHASE UNLOCKING	NO PHASE UNLOCKING
	INTERMEDIATE MVT	Brazilian Portuguese, Kipsigis, ...	English, Atchan, Nuntajiji, Tiwa, (a variety of) West Circassian, ...
	NO INTERMEDIATE MVT	Kalaallisut	Polish, Russian, Tsez

To give substance to this idea, we need to find independent correlates and/or explanations for whether a language has intermediate movement or not, and whether it has phase unlocking or not. That is the next step.

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